

HOTEL OPERATING SYSTEM (HOS)

Software Architecture, Development & UI/UX Design Specification

1. DOCUMENT PURPOSE

This document defines the technical architecture, software design, application structure, user hierarchy, module architecture, development standards, UI/UX principles, database strategy, API architecture, security model, integration architecture, deployment strategy, and development methodology for the Hotel Operating System (HOS).

The document is intended to serve as the primary technical reference for:

- UI/UX designers
- Frontend developers
- Backend developers
- Mobile developers
- Database developers
- QA engineers
- DevOps engineers
- Project managers
- Technical stakeholders

The system shall be designed as a **multi-tenant, cloud-native, modular SaaS platform** capable of supporting multiple hotel organizations and multiple properties.

2. SYSTEM VISION

The Hotel Operating System shall provide a centralized platform through which hotels can manage:

- Properties
- Rooms
- Reservations
- Guests
- Check-in/check-out
- Housekeeping
- Restaurant operations
- Events and banquets
- Inventory
- Billing
- Reports
- Channel Manager integrations
- AI Receptionist
- AI Lead Management
- Customer bookings
- Guest services

The platform shall support a hierarchical organization model:

Super Admin → Hotel Admin → Area Manager → Hotel Staff → Customer

3. APPLICATION ECOSYSTEM

The product shall consist of the following interfaces.

3.1 Super Admin Web Panel

Purpose: SaaS platform administration.

Users:

- Super Admin
- Platform Support

Responsibilities:

- Organization management
 - Hotel management
 - Platform configuration
 - Subscription management
 - System monitoring
 - Global integrations
 - Audit monitoring
-

3.2 Hotel Admin Web Panel

Purpose: Organization-level hotel administration.

Users:

- Hotel Admin
- Organization Administrator

Responsibilities:

- Hotels
 - Properties
 - Areas
 - Area Managers
 - Staff
 - Operations
 - Financial management
 - Reports
 - Integrations
-

3.3 Area Manager Panel

Purpose: Management of assigned geographical areas/hotels.

Users:

- Area Manager

Responsibilities:

- Assigned hotel monitoring
- Hotel performance
- Staff management
- Operational approvals
- Reservations
- Housekeeping
- Inventory
- CRM
- Reports

Access shall be restricted to assigned hotels.

3.4 Hotel Operations Web Panel

Purpose: Daily hotel operations.

Users:

- Hotel Manager
- Receptionist
- Reservation Executive
- Finance
- Restaurant Manager
- Inventory Manager
- Sales Manager

Modules:

- Front Desk
 - PMS
 - Reservations
 - Guests
 - Housekeeping
 - POS
 - Banquets
 - Inventory
 - Billing
 - CRM
 - Reports
-

3.5 Staff Mobile Application

Platform:

- Android
- iOS

Users:

- Reception
- Housekeeping
- Restaurant staff
- Inventory staff
- Maintenance staff
- Sales staff

The application shall dynamically display modules according to user permissions.

3.6 Customer Mobile Application

Platform:

- Android
- iOS

Functions:

- Hotel discovery
 - Room search
 - Booking
 - Online payment
 - Digital check-in
 - Stay management
 - Hotel services
 - Restaurant ordering
 - Room service
 - AI assistance
 - Notifications
 - Feedback
-

3.7 Customer Web Portal

A responsive customer-facing web application.

Functions:

- Hotel discovery
 - Availability
 - Booking
 - Payment
 - Reservation management
 - Digital check-in
 - Invoice
 - Guest services
 - Feedback
-

3.8 AI Receptionist Interface

The AI Receptionist shall be accessible through:

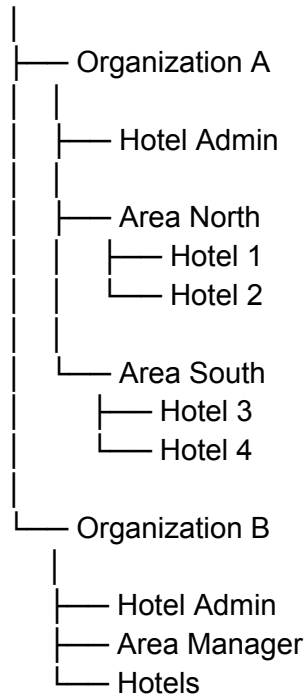
- Website
- Customer portal
- WhatsApp
- Voice/AI calling
- Future mobile application

Capabilities:

- FAQs
 - Availability lookup
 - Booking assistance
 - Reservation lookup
 - Guest information
 - Hotel information
 - Service requests
 - Human-agent escalation
-

4. ORGANIZATIONAL HIERARCHY

SUPER ADMIN



5. ACCESS CONTROL MODEL

The system shall use **RBAC + hierarchical resource-based authorization**.

Simple role checking is insufficient.

Authorization shall consider:

User
↓
Organization
↓
Area
↓
Property
↓
Department
↓
Resource
↓
Action

Example:

An Area Manager assigned to Hotel A shall not be able to access Hotel B, even if they manually modify the URL.

The backend must independently validate authorization.

6. CORE ROLES

Super Admin

Full platform access.

Hotel Admin

Full access to their organization.

Area Manager

Access only to assigned areas/hotels.

Hotel Manager

Access to assigned property.

Receptionist

Front-desk operations.

Reservation Executive

Reservations and guest booking operations.

Housekeeping Staff

Room cleaning and maintenance tasks.

Restaurant Staff

POS and restaurant operations.

Inventory Manager

Inventory and procurement.

Finance User

Billing, invoices and financial reports.

Sales/CRM User

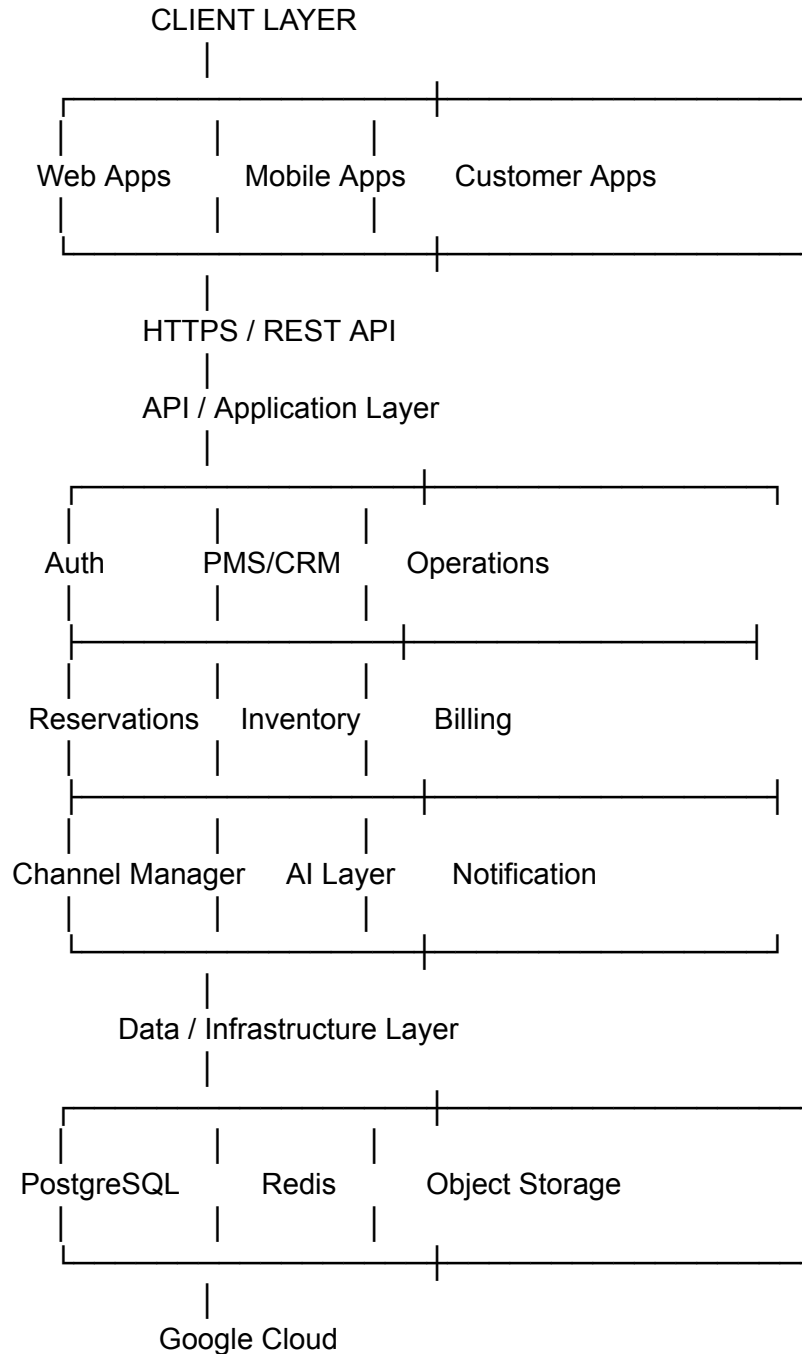
Leads, calls and follow-ups.

Customer

Guest-facing functionality only.

7. HIGH-LEVEL SYSTEM ARCHITECTURE

The recommended architecture is modular and cloud-ready.



8. RECOMMENDED TECHNOLOGY STACK

Frontend Web

- Next.js
- React
- TypeScript
- Tailwind CSS
- React Query
- Zod
- Form validation
- Component-based design system

Mobile

Flutter

Single codebase targeting:

- Android
- iOS

Backend

- Node.js
- NestJS
- TypeScript
- REST API
- WebSocket where required

Database

- PostgreSQL

Cache

- Redis

Storage

- Google Cloud Storage

Cloud

- Google Cloud Platform

Authentication

- JWT
- Refresh Tokens
- MFA/OTP
- Role-based authorization

AI

- OpenAI/Gemini or equivalent
- Function/tool calling
- AI knowledge base

AI Calling

- Vapi
- OmniDimension
- Equivalent API provider

Notifications

- Firebase Cloud Messaging
- Email provider
- WhatsApp Business API
- SMS provider

DevOps

- Docker
- Git
- GitHub/GitLab
- CI/CD

- Environment-based deployment
-

9. MULTI-TENANCY DESIGN

Every organization shall have a unique tenant identifier.

Example:

tenant_id

All organization-owned entities shall reference the tenant.

Example:

users
organizations
properties
rooms
reservations
guests
invoices
payments
leads
inventory

Each shall maintain tenant ownership.

The backend shall enforce tenant isolation at the service/repository layer.

10. CORE MODULE ARCHITECTURE

Module A — Authentication

- Registration
- Login
- OTP
- Password reset
- MFA
- Sessions
- Device management

Module B — Organization

- Organizations
- Admins
- Properties
- Areas
- Managers

Module C — PMS

- Rooms
- Room types
- Rates
- Availability
- Room status

Module D — Reservations

- Booking
- Walk-in
- Group booking
- Cancellation
- Modification
- Waitlist

Module E — Guest CRM

- Guest profile
- Guest history
- Preferences
- Feedback
- Complaints

Module F — Front Desk

- Arrivals
- Check-in
- In-house
- Check-out
- Stay extension

Module G — Housekeeping

- Room status
- Cleaning
- Tasks
- Maintenance
- Lost & found

Module H — POS

- Tables
- Orders
- KOT
- Billing
- Menu

Module I — Banquet

- Halls
- Events
- Packages
- Vendors
- Billing

Module J — Inventory

- Products
- Suppliers
- Purchase
- Stock
- Transfers

Module K — Billing

- Invoices
- Payments
- Refunds
- Taxes

Module L — Channel Manager

- Provider connection
- Mapping
- Availability sync
- Rates
- Reservations
- Restrictions
- Logs

Module M — AI Receptionist

- Knowledge base
- Conversations
- Availability
- Booking assistance
- FAQ

Module N — AI Lead Management

- Lead capture
 - Pipeline
 - Assignment
 - Calling
 - Recording
 - Transcription
 - Summary
 - Follow-up
 - Revenue analytics
-

11. DATABASE DESIGN PRINCIPLES

The database shall use normalized relational design.

Major entities:

Organization

Property

Area

User

Role

Permission

Building

Floor

RoomType

Room

RatePlan

Guest

Reservation

ReservationGuest

Stay

RoomAllocation

HousekeepingTask

MaintenanceRequest

Restaurant

Menu
MenuItem
Table
Order
OrderItem

BanquetHall
Event
EventPackage

Product
Supplier
PurchaseOrder
StockTransaction

Invoice
InvoiceItem
Payment
Refund

ChannelManager
ChannelMapping
SyncLog

Lead
LeadSource
LeadActivity
Call
CallRecording
FollowUp
LeadAssignment

Notification
AuditLog

12. API ARCHITECTURE

The API shall follow REST principles.

Example:

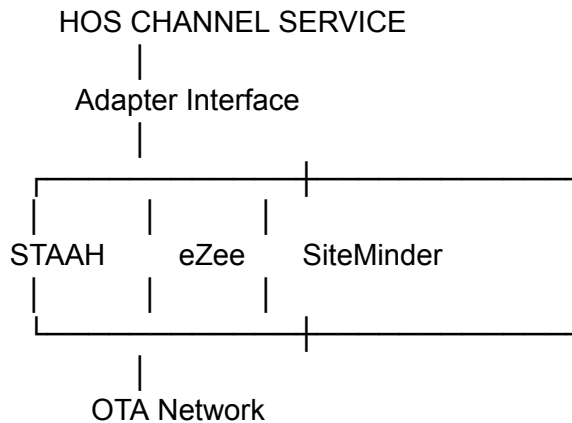
/api/v1/auth
/api/v1/organizations
/api/v1/properties
/api/v1/areas
/api/v1/users
/api/v1/roles
/api/v1/rooms
/api/v1/reservations
/api/v1/guests
/api/v1/checkins
/api/v1/checkouts
/api/v1/housekeeping
/api/v1/pos
/api/v1/inventory
/api/v1/invoices
/api/v1/payments
/api/v1/channel-managers
/api/v1/ai/receptionist
/api/v1/leads
/api/v1/calls
/api/v1/followups
/api/v1/reports
/api/v1/notifications
/api/v1/audit-logs

Each endpoint shall define:

- Authentication
 - Authorization
 - Request schema
 - Response schema
 - Validation
 - Error codes
 - Audit requirements
-

13. CHANNEL MANAGER ARCHITECTURE

The system shall use an adapter-based integration architecture.



Each provider shall have its own adapter.

This prevents provider-specific logic from contaminating the PMS core.

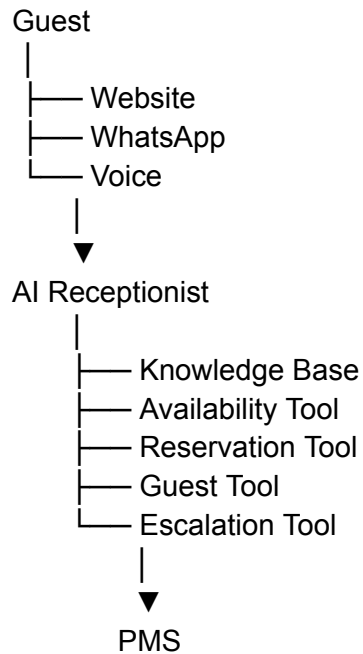
The system shall support:

- Reservation synchronization
- Availability synchronization
- Rate synchronization
- Restriction synchronization
- Cancellation synchronization
- Mapping
- Error handling
- Retry mechanisms
- Sync logs

For hotels without a Channel Manager:

Manual Inventory Management shall be available.

14. AI RECEPTIONIST ARCHITECTURE



The AI shall not directly modify critical hotel data without controlled backend tools and authorization.

15. AI LEAD MANAGEMENT ARCHITECTURE

Lead Source

- Website
- WhatsApp
- Phone
- Social
- Referral



Lead Capture



AI Qualification



Assignment



AI/Staff Call



Recording + Transcript



AI Summary



Follow-Up



Reservation



Revenue



16. UI/UX DESIGN PRINCIPLES

The system shall follow enterprise dashboard principles.

Design Requirements

- Clean interface
 - Consistent spacing
 - Clear hierarchy
 - Responsive design
 - Accessible typography
 - Consistent component library
 - Minimal unnecessary animations
 - Clear status indicators
 - Confirmation before destructive actions
 - Loading states
 - Empty states
 - Error states
-

17. SCREEN DESIGN STANDARD

Every screen specification shall include:

Header

- Page title
- Breadcrumb
- Search
- Notifications
- User menu

Primary Content

- Cards
- Tables
- Forms
- Charts
- Filters
- Tabs

Actions

- Create
- Edit
- Delete
- Export
- Print
- Approve
- Cancel

System States

- Loading
 - Empty
 - Error
 - Success
 - Permission denied
-

18. SECURITY ARCHITECTURE

Mandatory controls:

- HTTPS
 - Password hashing
 - JWT
 - Refresh token rotation
 - MFA
 - RBAC
 - Tenant isolation
 - Resource-level authorization
 - API rate limiting
 - Input validation
 - SQL injection protection
 - XSS protection
 - CSRF protection where applicable
 - Secure headers
 - Audit logging
 - Secrets management
 - Database backups
-

19. AUDIT LOGGING

Important actions shall be logged.

Example:

User:

Neeraj

Action:

UPDATE_RESERVATION

Resource:

Reservation #RES-10293

Old Value:

Room 201

New Value:

Room 305

Timestamp:

2026-08-11 14:32

IP:

Recorded

Device:

Recorded

20. ERROR HANDLING

The API shall use standardized error responses.

Example:

```
{
  "success": false,
  "error": {
    "code": "RESERVATION_NOT_FOUND",
    "message": "Reservation could not be found."
  },
  "requestId": "REQ-123456"
}
```

Errors shall be logged with a request ID for troubleshooting.

21. ENVIRONMENT ARCHITECTURE

Three primary environments are recommended:

Development



Staging / UAT



Production

Each environment shall have separate:

- Database
- Environment variables
- API credentials
- Storage
- Logs

Production credentials must never be committed to source control.

22. DEVELOPMENT WORKFLOW

The development process shall follow:

Requirement



Technical Design



UI/UX



Database



API



Frontend



Integration



Testing



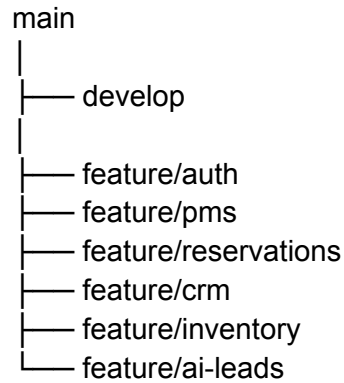
UAT



Production

23. GIT BRANCHING

Recommended:



Pull requests shall be reviewed before merging.

24. TESTING STRATEGY

Testing shall include:

Unit Testing

Business logic and services.

API Testing

- Authentication
- Validation
- Authorization
- Response correctness

Integration Testing

- PMS
- Channel Manager
- AI
- Payments
- Notifications

UI Testing

- Forms
- Navigation
- Responsive behavior
- Role-based visibility

Security Testing

- Tenant isolation
- RBAC
- Authentication
- API authorization

UAT

Hotel users shall validate operational workflows before production release.

25. PERFORMANCE REQUIREMENTS

The application should be designed for:

- Efficient database queries
- Pagination
- Caching
- Lazy loading
- Background jobs
- Queue-based integrations
- Optimized API responses

Long-running operations should not block normal API requests.

Examples:

- Channel synchronization
- AI processing
- Email sending
- Report generation
- Call processing

These should preferably execute asynchronously.

26. OBSERVABILITY

The production system should provide:

- Application logs
- API logs
- Error tracking
- Database monitoring
- Server monitoring
- Health checks
- Integration monitoring
- Alerting

Critical services shall expose health endpoints.

27. BACKUP & RECOVERY

Database backups shall be configured according to the selected GCP architecture.

The system should support:

- Automated backups
 - Point-in-time recovery where supported
 - Storage backups
 - Disaster recovery procedures
-

28. V1 DEVELOPMENT PRINCIPLE

The system shall be architected for the complete product vision but development shall be divided into:

V1 — Core Production MVP

Priority:

1. Authentication
2. Multi-tenancy
3. Hotel/property management
4. RBAC
5. PMS
6. Reservations
7. Guest management
8. Check-in/out
9. Housekeeping
10. Billing
11. Inventory
12. Manual inventory
13. Core reports
14. Channel Manager framework
15. AI Receptionist foundation
16. AI Lead Management foundation
17. Admin/Manager dashboards
18. Staff application foundation
19. Customer booking interface
20. Production deployment

Future Enhancements

Potential future scope:

- Multiple advanced OTA integrations
 - Advanced revenue management
 - Advanced loyalty engine
 - Smart locks
 - IoT
 - Facial recognition
 - Advanced voice AI
 - Predictive analytics
 - Advanced accounting
 - White-label native hotel apps
 - Enterprise BI
-

29. DEVELOPMENT GOVERNANCE

No module should be developed without:

1. Approved requirement
2. UI/UX specification
3. Database design
4. API specification
5. Permission definition
6. Validation rules
7. Error states
8. Acceptance criteria

This prevents the common problem where frontend, backend and database teams implement different interpretations of the same feature.

30. DEFINITION OF DONE

A feature shall be considered complete only when:

- UI implemented
 - API implemented
 - Database migration completed
 - Validation implemented
 - Permission checks implemented
 - Error handling implemented
 - Audit requirements implemented
 - Responsive behavior verified
 - Unit/API testing completed
 - Integration testing completed where applicable
 - Code reviewed
 - Documentation updated
-

31. MASTER PRODUCT STRUCTURE

The final system shall therefore consist of:

Web

1. Super Admin Panel
2. Hotel Admin Panel
3. Area Manager Panel
4. Hotel Operations Panel
5. Customer Web Portal

Mobile

6. Staff Mobile Application
7. Customer Mobile Application

AI / Communication

8. AI Receptionist Interface
9. AI Calling Agent Integration

These interfaces shall share the same secure backend and centralized identity/authorization system while maintaining strict separation of user permissions and data access.

32. ARCHITECTURAL PRINCIPLE

The central principle of the system is:

One platform, one source of truth for hotel operations, strict tenant isolation, hierarchical access control, modular services, and API-first integration.

The system should be built so that adding another hotel, property, manager, Channel Manager, OTA, AI provider, or mobile client does not require rewriting the core PMS.

This architecture provides the foundation for evolving the product from an initial hotel management system into a scalable hospitality SaaS platform.

#	Interface	Platform	Users
1	Super Admin Panel	Web	Super Admin
2	Hotel Admin Panel	Web	Hotel Admin
3	Area Manager Panel	Web	Area Managers
4	Hotel Operations Panel	Web	Hotel Managers & Staff
5	Staff App	Android/iOS	Hotel Employees
6	Customer App	Android/iOS	Hotel Guests
7	Customer Web Portal	Web	Hotel Guests
8	AI Receptionist Interface	Web + WhatsApp + Voice	Guests/Customers

1. PRODUCT OVERVIEW

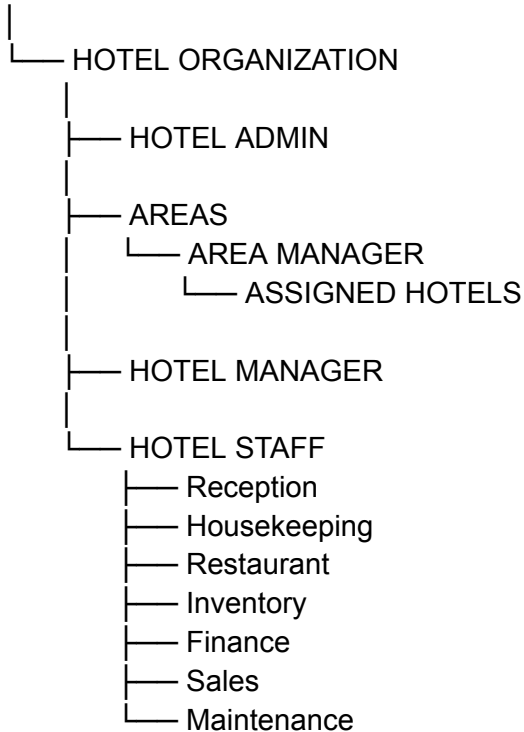
The Hotel Operating System (HOS) is a cloud-based, multi-tenant hospitality management platform designed to manage one or multiple hotels/properties from a centralized system.

The platform covers:

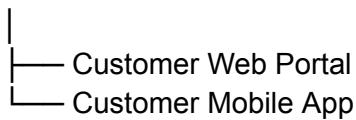
- SaaS administration
 - Hotel/property management
 - PMS
 - Reservations
 - Walk-in bookings
 - Front desk
 - Guest CRM
 - Housekeeping
 - Restaurant POS
 - Banquet/event management
 - Billing and invoicing
 - Inventory
 - Channel Manager integration
 - Manual inventory management
 - AI Receptionist
 - AI Lead Management
 - Customer booking
 - Staff mobile operations
 - Reports and analytics
 - Notifications
 - Audit and security
-

2. PLATFORM USER HIERARCHY

SUPER ADMIN



CUSTOMER



3. FEATURE PRIORITY

Each feature can be classified as:

P0 — Core Production Requirement

Required for the initial production system.

P1 — Important Business Feature

Required for a complete operational platform.

P2 — Advanced/Future Enhancement

Can be expanded after the initial production release.

4. SUPER ADMIN MODULE

4.1 Super Admin Authentication

4.1.1 Admin Login

Allows the platform administrator to securely access the Super Admin panel.

Use: Prevent unauthorized access to the SaaS management platform.

4.1.2 MFA/OTP

Adds a second authentication layer.

Use: Protects privileged administrator accounts.

4.1.3 Forgot Password

Allows secure password recovery.

4.1.4 Session Management

Displays active sessions/devices.

Use: Allows administrators to terminate suspicious sessions.

4.1.5 Admin Logout

Securely terminates the session.

5. SUPER ADMIN DASHBOARD

5.1 Platform Overview

Displays:

- Total organizations
- Total hotels
- Total properties
- Total users
- Active users
- Active subscriptions
- Total reservations
- Total revenue
- Active integrations
- System alerts

Use: Gives the Super Admin a real-time overview of the entire SaaS platform.

5.2 Hotel Statistics

Displays hotel-wise:

- Occupancy
- Reservations
- Revenue
- Cancellations
- Active rooms

Use: Monitor platform-wide hotel performance.

5.3 Revenue Analytics

Displays:

- Monthly revenue
- Organization-wise revenue
- Subscription revenue
- Growth rate

Use: Understand SaaS business performance.

5.4 System Health

Displays:

- API status
- Database status
- Queue status
- Integration status
- Error rate

Use: Identify technical issues before they affect customers.

6. ORGANIZATION MANAGEMENT

6.1 Organization List

Displays:

- Organization name
- Owner
- Number of hotels
- Users
- Subscription
- Status
- Created date

Use: Manage all hotel organizations registered on the SaaS platform.

6.2 Create Organization

Fields:

- Organization name
- Legal name
- Owner
- Email
- Phone
- Address
- GST/tax information
- Logo
- Subscription
- Limits

Use: Create a new hotel organization/tenant.

6.3 Organization Details

Tabs:

- Overview
- Hotels
- Users
- Admins
- Subscription
- Billing
- Usage
- Integrations
- Logs

6.4 Suspend Organization

Temporarily disables access.

Use: Handle payment issues, policy violations, or administrative suspension.

6.5 Activate Organization

Restores access.

7. HOTEL ADMINISTRATION

7.1 Hotel List

Displays:

- Hotel name
- Location
- Manager
- Rooms
- Occupancy
- Status

7.2 Add Hotel

Fields:

- Hotel name
- Address
- Contact
- Email
- Property type
- Room count
- Check-in time
- Check-out time
- Tax information
- Facilities

Use: Register a new property under an organization.

7.3 Hotel Profile

Contains:

- General information
- Contact information
- Logo
- Photos
- Facilities
- Policies
- Tax details
- Payment settings

7.4 Property Configuration

Allows configuration of:

- Floors
 - Buildings
 - Room types
 - Rooms
 - Amenities
 - Rates
-

8. AREA MANAGEMENT

8.1 Create Area

Examples:

- North Area
- South Area
- City Center
- District A

Use: Group hotels geographically or operationally.

8.2 Area List

Displays:

- Area
- Manager
- Hotels
- Rooms
- Occupancy
- Revenue

8.3 Area Manager Assignment

Allows Hotel Admin to assign a manager.

8.4 Hotel Assignment

Allows multiple hotels to be assigned to an Area Manager.

8.5 Area Performance

Displays:

- Occupancy
 - Revenue
 - Reservations
 - Leads
 - Staff performance
-

9. AREA MANAGER MODULE

9.1 Area Dashboard

Displays:

- Assigned hotels
- Occupancy
- Revenue
- Arrivals
- Departures
- Housekeeping
- Leads
- Staff
- Alerts

Use: Monitor all assigned properties from one dashboard.

9.2 Assigned Hotels

Shows only hotels authorized for the manager.

9.3 Hotel Comparison

Compare:

- Revenue
- Occupancy
- ADR
- RevPAR
- Reservations
- Cancellations

9.4 Staff Monitoring

View:

- Staff attendance
- Tasks
- Performance
- Department activity

9.5 Approval Center

Manager can approve:

- Discounts
 - Refunds
 - Purchase requests
 - Stock adjustments
 - Maintenance requests
-

10. USER & ROLE MANAGEMENT

10.1 User List

Displays:

- Name
- Email
- Phone
- Role
- Hotel
- Area
- Status

10.2 Create User

Fields:

- Name
- Email
- Phone
- Role
- Department
- Hotel
- Area
- Permissions

10.3 Role Management

Create and manage roles.

10.4 Permission Management

Permissions can include:

- View
- Create
- Edit
- Delete
- Approve
- Export
- Refund
- Manage

10.5 User Assignment

Assign user to:

- Organization
- Area
- Hotel
- Department

10.6 Account Suspension

Temporarily disable users.

11. AUTHENTICATION & SECURITY

11.1 Email Login

Secure email/password authentication.

11.2 Mobile OTP

Login through OTP.

11.3 MFA

Additional authentication for privileged accounts.

11.4 Password Management

- Change password
- Reset password
- Password policy

11.5 Device Management

Shows trusted/active devices.

11.6 Login History

Records:

- Login time
- Device
- IP
- Location where available

11.7 Session Control

Force logout from other devices.

12. PMS — PROPERTY MANAGEMENT SYSTEM

The PMS is the central operational module.

12.1 Property Dashboard

Displays:

- Occupancy
- Available rooms
- Occupied rooms
- Dirty rooms
- Out-of-order rooms
- Arrivals
- Departures
- Revenue

12.2 Building Management

Create multiple buildings.

Use: Supports large properties or hotel groups.

12.3 Floor Management

Create floors and assign rooms.

12.4 Room Type Management

Examples:

- Standard
- Deluxe
- Suite
- Executive

Information:

- Capacity
- Bed type
- Amenities
- Base price
- Images

12.5 Room Management

Fields:

- Room number
- Room type
- Floor
- Status
- Amenities
- Housekeeping status

12.6 Room Status

Statuses:

- Available
- Occupied
- Reserved
- Dirty
- Cleaning
- Clean
- Inspected
- Out of Order
- Maintenance

12.7 Room Map

Visual room layout.

Use: Front desk can quickly understand room availability.

12.8 Room Availability

Displays room availability by date.

12.9 Rate Plans

Create:

- Standard rate
- Corporate rate
- Weekend rate
- Seasonal rate

12.10 Restrictions

Configure:

- Minimum stay
 - Maximum stay
 - Closed dates
 - Closed room types
-

13. RESERVATION MANAGEMENT

13.1 Reservation Dashboard

Displays:

- Today's bookings
- Upcoming bookings
- Cancellations
- Pending payments
- No-shows

13.2 Reservation Search

Search by:

- Booking ID
- Guest
- Phone
- Email
- Room
- Date

13.3 Create Reservation

Fields:

- Guest
- Room type
- Room
- Check-in
- Check-out
- Guests
- Rate
- Discount
- Taxes
- Payment

13.4 Walk-In Booking

Allows reception to create an immediate reservation without prior booking.

13.5 Group Booking

Allows multiple rooms to be booked under one group.

13.6 Booking Modification

Change:

- Dates
- Room
- Guest
- Rate
- Number of guests

13.7 Cancellation

Cancel reservation with:

- Cancellation reason
- Refund calculation
- Cancellation policy

13.8 No-Show

Mark reservation as no-show.

13.9 Waitlist

Maintain guests waiting for rooms.

13.10 Reservation Calendar

Calendar view of bookings.

13.11 Booking Confirmation

Generate confirmation document/email/message.

14. FRONT DESK

14.1 Arrival List

Shows today's expected arrivals.

14.2 Departure List

Shows today's expected departures.

14.3 Check-In

Process:

1. Reservation verification
2. Guest verification
3. ID capture
4. Room assignment
5. Payment/deposit
6. Registration
7. Check-in

14.4 Walk-In Check-In

Direct check-in without existing reservation.

14.5 Room Assignment

Assign available room.

14.6 Room Change

Move guest to another room.

14.7 Stay Extension

Extend checkout date.

14.8 Check-Out

Process:

- Final bill
- Additional charges
- Payment
- Invoice
- Room release

14.9 Early Check-In

Handle guests arriving before standard check-in time.

14.10 Late Check-Out

Handle guests leaving after standard checkout time.

15. GUEST CRM

15.1 Guest Profile

Contains:

- Name
- Phone
- Email
- Address
- ID
- Preferences

15.2 Guest History

Shows:

- Previous bookings
- Stays
- Payments
- Complaints
- Feedback

15.3 Guest Preferences

Examples:

- Room preference
- Food preference
- Special requests

15.4 Guest Documents

Store permitted guest documents securely.

15.5 Guest Segmentation

Examples:

- VIP
- Corporate
- Repeat guest
- New guest

15.6 Guest Feedback

Collect ratings and comments.

15.7 Complaint Management

Record:

- Complaint
 - Department
 - Priority
 - Assigned staff
 - Resolution
 - Status
-

16. HOUSEKEEPING

16.1 Housekeeping Dashboard

Shows:

- Dirty rooms
- Clean rooms
- Cleaning
- Inspection pending
- Maintenance

16.2 Room Board

Visual representation of room status.

16.3 Cleaning Task

Create task with:

- Room
- Staff
- Priority
- Deadline

16.4 Task Assignment

Assign rooms to housekeeping staff.

16.5 Cleaning Checklist

Examples:

- Bed
- Bathroom
- Towels
- Amenities
- Floor
- Trash
- Inspection

16.6 Room Inspection

Supervisor verifies room quality.

16.7 Maintenance Request

Staff can report:

- AC issue
- Plumbing
- Electrical
- Furniture

16.8 Lost & Found

Record:

- Item
 - Room
 - Date
 - Description
 - Photo
 - Status
-

17. RESTAURANT POS

17.1 Restaurant Setup

Configure:

- Restaurant
- Sections
- Tables
- Menu

17.2 Table Management

Displays:

- Available
- Occupied
- Reserved
- Cleaning

17.3 Menu Management

Manage:

- Categories
- Items
- Prices
- Taxes
- Add-ons

17.4 POS

Features:

- Add item
- Quantity
- Discount
- Tax
- Notes
- Payment

17.5 KOT

Kitchen Order Ticket.

Use: Sends order details to kitchen staff.

17.6 Order Tracking

Statuses:

- New
- Preparing
- Ready
- Served
- Completed

17.7 Room Charge

Restaurant charges can be added to the guest room bill.

17.8 Restaurant Billing

Generate restaurant invoices.

17.9 Restaurant Reports

- Sales
 - Items
 - Categories
 - Tax
 - Staff performance
-

18. BANQUET & EVENT MANAGEMENT

18.1 Hall Management

Create halls with:

- Capacity
- Layout
- Facilities
- Pricing

18.2 Event Calendar

View upcoming events.

18.3 Event Booking

Fields:

- Customer
- Event type
- Date
- Time
- Hall
- Guests
- Package
- Price

18.4 Event Packages

Create packages containing:

- Food
- Decoration
- Hall
- Equipment
- Services

18.5 Event Billing

Generate event invoice.

18.6 Event Staff Assignment

Assign staff to events.

19. INVENTORY MANAGEMENT

19.1 Inventory Dashboard

Displays:

- Total products
- Current stock
- Low stock
- Purchase orders
- Stock value

19.2 Product Management

Fields:

- Product
- SKU
- Category
- Unit
- Price
- Minimum stock

19.3 Supplier Management

Store:

- Supplier
- Contact
- Address
- Products
- Payment terms

19.4 Purchase Request

Staff requests stock.

19.5 Purchase Order

Create supplier purchase orders.

19.6 Goods Receipt

Record received products.

19.7 Stock Adjustment

Adjust stock for:

- Damage
- Loss
- Correction
- Expiry

19.8 Stock Transfer

Transfer stock between locations.

19.9 Low Stock Alerts

Automatically notify responsible staff.

19.10 Stock Ledger

Maintain all stock movements.

20. BILLING & INVOICING

20.1 Invoice Management

Create/manage invoices.

20.2 Invoice Items

Include:

- Room
- Restaurant
- Services
- Taxes
- Discounts

20.3 Payment Management

Support configured payment methods.

20.4 Partial Payment

Allow partial payment.

20.5 Refund

Process approved refunds.

20.6 Tax Management

Configure applicable taxes.

20.7 Invoice PDF

Generate downloadable/printable invoice.

20.8 Payment History

Track all transactions.

21. CHANNEL MANAGER INTEGRATION

The system shall support a generic Channel Manager architecture.

Supported providers may include:

- STAAH
- eZee
- SiteMinder
- Hotelogix
- Other API-based providers

21.1 Provider Configuration

Store provider credentials securely.

21.2 Property Mapping

Map:

HOS Room Type



Channel Manager Room Type



OTA Room Type

21.3 Rate Mapping

Map HOS rate plans to channel rates.

21.4 Availability Sync

Synchronize available inventory.

21.5 Reservation Sync

Import reservations from channels.

21.6 Rate Sync

Send hotel rates.

21.7 Restriction Sync

Synchronize:

- Minimum stay
- Maximum stay
- Closed dates
- Stop sell

21.8 Cancellation Sync

Receive reservation cancellation information.

21.9 Sync Logs

Track:

- Request
- Response
- Timestamp
- Status
- Error

21.10 Retry Mechanism

Retry failed synchronization.

22. MANUAL INVENTORY MANAGEMENT

This feature is mandatory for hotels without a Channel Manager.

22.1 Manual Availability

Staff can manually update availability.

22.2 Manual Rate Update

Update room prices.

22.3 Manual Restrictions

Set:

- Closed room
- Stop sell
- Minimum stay

22.4 Inventory Calendar

Visual calendar for availability.

22.5 Manual Reservation Entry

Staff can enter bookings received through:

- Phone
 - Walk-in
 - Direct enquiry
 - Offline channels
-

23. AI RECEPTIONIST

23.1 AI Chat

Guest can ask questions.

23.2 Hotel FAQ

AI answers:

- Check-in time
- Check-out time
- Facilities
- Restaurant timing
- Policies

23.3 Live Availability

AI can query current availability.

23.4 Booking Assistance

AI guides guest through booking.

23.5 Reservation Lookup

Guest can ask about an existing booking.

23.6 Guest Information

AI can retrieve authorized information.

23.7 Service Requests

Guest can request:

- Towels
- Housekeeping
- Restaurant
- Maintenance

23.8 Human Escalation

Transfer conversation to hotel staff.

23.9 AI Knowledge Base

Hotel Admin can manage:

- FAQs
- Policies
- Hotel information
- Services
- Rules

23.10 AI Conversation History

Authorized users can view conversations.

24. AI LEAD MANAGEMENT

This is a separate business-growth module.

24.1 Lead Capture

Capture leads from:

- Website
- Customer app
- Phone
- WhatsApp
- Social channels
- Referral
- Manual entry

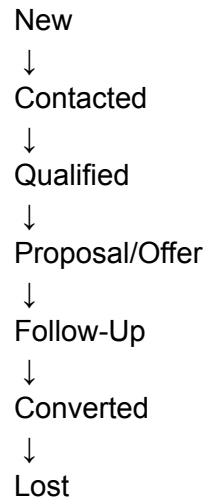
24.2 Lead Profile

Contains:

- Name
- Phone
- Email
- Source
- Requirement
- Expected stay
- Budget
- Lead value

24.3 Lead Pipeline

Example:



24.4 Lead Assignment

Assign leads to sales staff.

Assignment can be:

- Manual
- Round-robin
- Area-based
- Hotel-based

24.5 AI Calling Agent

Integration with:

- Vapi
- OmniDimension
- Equivalent provider

The AI calling agent can:

- Call leads
- Introduce hotel
- Understand requirements
- Answer basic questions
- Qualify leads
- Schedule follow-up

24.6 Call Recording

Store call recording where legally and technically permitted.

24.7 Call Transcription

Convert conversation into text.

24.8 AI Call Summary

Generate:

- Customer requirement
- Interest level
- Budget
- Preferred dates
- Next action

24.9 Auto Follow-Up

Automatically schedule:

- Call
- WhatsApp
- Email
- Reminder

24.10 Lead Analytics

Track:

- Leads
- Conversion
- Lost leads
- Response time
- Sales performance

24.11 Revenue Source Analytics

Identify revenue by:

- Website
- OTA
- Direct
- WhatsApp
- Phone
- Referral
- Sales staff

24.12 Lead Assignment Analytics

Track performance by:

- Employee
 - Hotel
 - Area
 - Source
-

25. CUSTOMER MOBILE APP

25.1 Customer Registration

- Name
- Phone
- Email
- Password/OTP

25.2 Customer Profile

- Personal information
- Preferences
- Booking history

25.3 Hotel Discovery

Search:

- Location
- Hotel
- Date
- Guests

25.4 Hotel Details

Display:

- Photos
- Rooms
- Amenities
- Facilities
- Policies
- Reviews

25.5 Room Search

Filters:

- Price
- Room type
- Capacity
- Amenities

25.6 Booking

Process:

Search



Select Room



Guest Details



Review Price



Payment



Confirmation

25.7 My Bookings

Display:

- Upcoming
- Current
- Previous
- Cancelled

25.8 Digital Check-In

Guest can submit information before arrival.

25.9 Hotel Services

Guest can request:

- Housekeeping
- Towels
- Laundry
- Maintenance
- Concierge

25.10 Restaurant

Guest can:

- Browse menu
- Order food
- Track order
- Pay

25.11 Notifications

- Booking confirmation
- Check-in reminder
- Checkout reminder
- Service updates
- Offers

25.12 AI Assistant

Customer can communicate with AI Receptionist.

25.13 Reviews

Submit rating and feedback.

26. CUSTOMER WEB PORTAL

The web portal shall provide core customer functionality without requiring app installation.

Features:

- Registration
 - Login
 - Hotel search
 - Room availability
 - Booking
 - Payment
 - Booking management
 - Digital check-in
 - Invoice
 - Service requests
 - AI Receptionist
 - Feedback
-

27. STAFF MOBILE APP

One application shall serve different staff roles through permissions.

27.1 Staff Dashboard

Dashboard is role-specific.

27.2 Reception

- Arrivals
- Departures
- Reservations
- Check-in
- Check-out

27.3 Housekeeping

- Assigned rooms
- Cleaning
- Room status
- Maintenance

27.4 Restaurant

- Orders
- KOT
- Tables
- Billing

27.5 Inventory

- Stock
- Requests
- Transfers
- Goods receipt

27.6 Sales

- Leads
- Calls
- Follow-ups
- Customer information

27.7 Notifications

Push notifications for assigned tasks.

27.8 Task Management

Staff can:

- Accept task
- Start task
- Complete task
- Add notes
- Upload photos

28. NOTIFICATION SYSTEM

28.1 Push Notifications

For mobile applications.

28.2 Email Notifications

For:

- Booking
- Invoice
- Account
- Alerts

28.3 SMS

For important transactional messages where configured.

28.4 WhatsApp

For:

- Booking confirmation
- Reminders
- Customer communication

28.5 Internal Notifications

For staff:

- New task
 - New lead
 - Approval
 - Reservation
 - Maintenance
-

29. REPORTING & ANALYTICS

29.1 Executive Dashboard

High-level:

- Revenue
- Occupancy
- ADR
- RevPAR
- Bookings

29.2 Reservation Reports

- Booking source
- Booking date
- Stay date
- Cancellation
- No-show

29.3 Occupancy Report

- Daily
- Weekly
- Monthly
- Property-wise

29.4 Revenue Report

- Room revenue
- Restaurant revenue
- Banquet revenue
- Service revenue

29.5 Guest Report

- New guests
- Repeat guests
- VIP guests
- Feedback

29.6 Housekeeping Report

- Cleaning performance
- Pending tasks
- Maintenance

29.7 Inventory Report

- Stock
- Purchase
- Consumption
- Low stock

29.8 POS Report

- Sales
- Items
- Categories
- Taxes

29.9 Lead Report

- Lead source
- Conversion
- Revenue
- Salesperson performance

29.10 Manager Reports

Area Managers can compare assigned hotels.

30. SEARCH & FILTERING

Global search where appropriate.

Search across:

- Reservations
- Guests
- Rooms
- Leads
- Inventory
- Invoices

Filters:

- Date
 - Hotel
 - Area
 - Status
 - User
 - Source
-

31. EXPORT & DOCUMENTS

31.1 PDF Export

For:

- Invoices
- Reports
- Reservations

31.2 Excel/CSV Export

For authorized reports and datasets.

31.3 Print

For:

- Invoice
 - Reservation
 - KOT
 - Reports
-

32. AUDIT LOG

Every critical action should be traceable.

Examples:

- User login
- Reservation modification
- Room change
- Invoice modification
- Refund
- Stock adjustment
- User creation
- Permission change
- Channel sync

Audit record:

User

Action

Module

Resource

Old Value

New Value

Timestamp

IP

Device

33. SETTINGS MODULE

33.1 Organization Settings

- Organization details
- Logo
- Contact

33.2 Hotel Settings

- Check-in time
- Checkout time
- Policies
- Taxes

33.3 User Settings

- Roles
- Permissions

33.4 Notification Settings

Configure notification channels.

33.5 Payment Settings

Configure payment methods.

33.6 Integration Settings

Configure:

- Channel Managers
- AI providers
- Payment gateway
- WhatsApp
- Email

33.7 Security Settings

- MFA
 - Password policy
 - Sessions
 - Access controls
-

34. API & INTEGRATION MANAGEMENT

34.1 API Keys

Generate/manage API credentials where required.

34.2 Webhooks

Support incoming/outgoing event notifications.

34.3 Integration Logs

Track API activity.

34.4 API Error Monitoring

Track failed requests.

34.5 Retry Management

Retry recoverable integration failures.

35. SYSTEM ADMINISTRATION

35.1 System Health

Monitor:

- API
- Database
- Storage
- Queue
- Integrations

35.2 Error Monitoring

Track application errors.

35.3 Background Jobs

Monitor:

- Notifications
- AI jobs
- Channel sync
- Reports

35.4 Backup Monitoring

Verify backup status.

36. SECURITY FEATURES

36.1 Tenant Isolation

Organization A cannot access Organization B data.

36.2 Resource-Level Authorization

User can only access assigned hotels/resources.

36.3 Encryption

Sensitive data should be protected in transit and appropriately at rest.

36.4 Password Hashing

Passwords must never be stored as plaintext.

36.5 Rate Limiting

Protect APIs against abuse.

36.6 Input Validation

Validate all API requests.

36.7 Secure File Upload

Restrict file types and sizes.

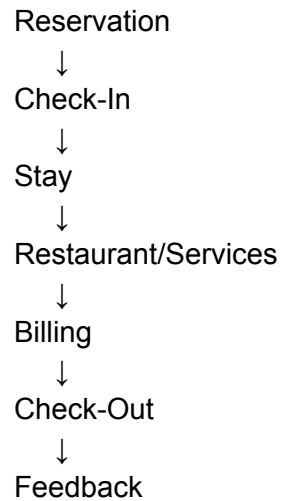
36.8 Audit Trail

Maintain critical activity history.

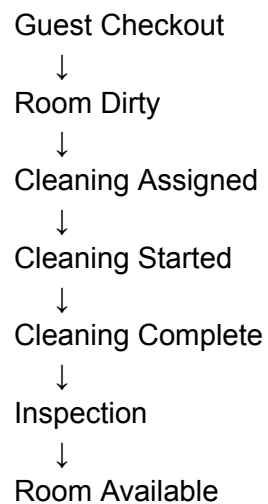
37. WORKFLOW ENGINE

The system should support configurable operational workflows.

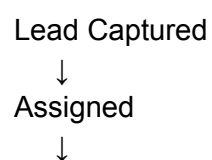
Example:



For housekeeping:



For leads:



AI/Staff Contact



Qualified



Follow-Up



Booking



Revenue

38. CORE BUSINESS RULES

38.1 Reservation Rule

A room cannot be double-booked for overlapping confirmed stays unless explicitly configured for the property.

38.2 Availability Rule

Availability must consider:

- Confirmed reservations
- Blocked rooms
- Out-of-order rooms
- Manual inventory
- Channel inventory

38.3 Channel Manager Rule

Where a Channel Manager is configured, synchronization must follow the configured integration strategy.

38.4 Manual Inventory Rule

Hotels without a Channel Manager shall manage availability manually.

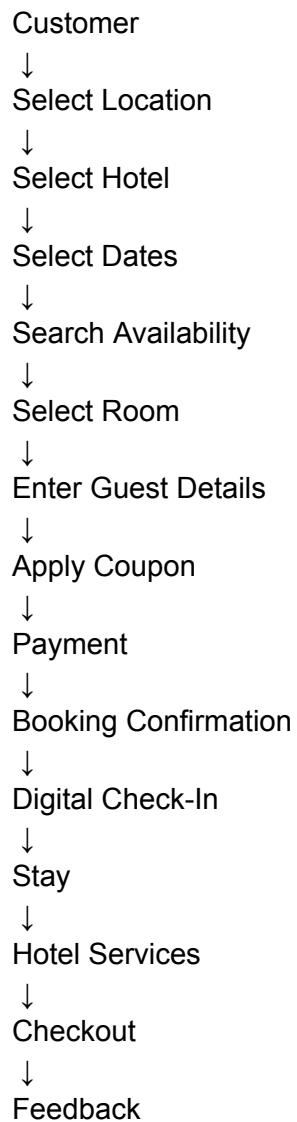
38.5 Access Rule

No user can access a hotel outside their authorized scope.

38.6 Financial Rule

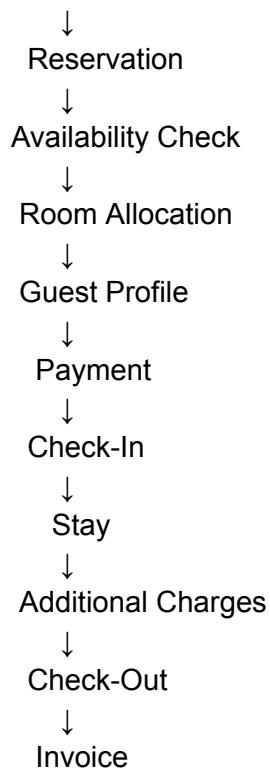
Refunds and financial adjustments may require approval based on role.

39. CUSTOMER BOOKING WORKFLOW



40. HOTEL RESERVATION WORKFLOW

OTA / Website / Phone / Walk-In



41. DEVELOPMENT MODULE PRIORITY

Phase 1 — Foundation

- Authentication
- User management
- Roles
- Permissions
- Organization
- Hotel
- Area
- Multi-tenancy
- Database foundation

Phase 2 — PMS

- Rooms
- Room types
- Availability
- Reservations
- Walk-in
- Check-in
- Check-out
- Guest CRM

Phase 3 — Operations

- Housekeeping
- Inventory
- Billing
- POS
- Basic banquet

Phase 4 — Integrations

- Channel Manager framework
- Manual inventory
- Notifications
- Payment integration

Phase 5 — AI

- AI Receptionist
- Lead capture
- Lead pipeline
- AI calling integration
- Call summary
- Follow-up

Phase 6 — Customer

- Customer web
- Customer mobile
- Booking
- Digital check-in
- Hotel services

Phase 7 — QA & Production

- Security testing
 - API testing
 - UI testing
 - Integration testing
 - UAT
 - Deployment
 - Monitoring
-

42. FEATURE DEPENDENCY MODEL

The development order is important.

Authentication



Multi-Tenancy



RBAC



Organization



Hotels / Areas



Rooms



Availability



Reservations



Guests



Check-In / Check-Out



Billing



Housekeeping / POS / Inventory



Channel Manager



AI



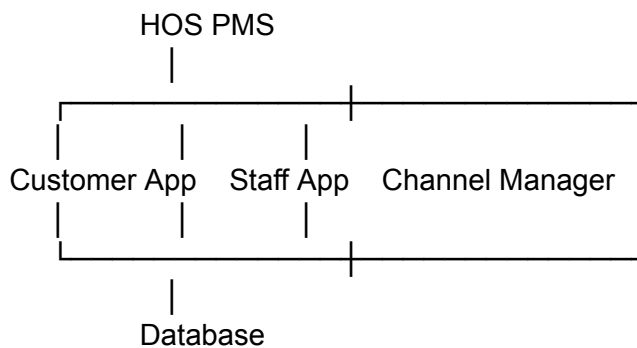
Customer Applications

43. IMPORTANT ARCHITECTURAL DECISION

The PMS shall remain the central operational system.

The architecture should avoid allowing external systems to independently maintain conflicting hotel inventory without synchronization rules.

Conceptually:



For Channel Manager integrations, provider-specific adapters shall isolate external API differences from the PMS core.

44. MASTER FEATURE COUNT

For planning purposes, the platform contains approximately **40+ major modules/categories** and **300+ functional sub-features/workflows** when all operational, administrative, integration, security, reporting and customer capabilities are decomposed into individual development items.

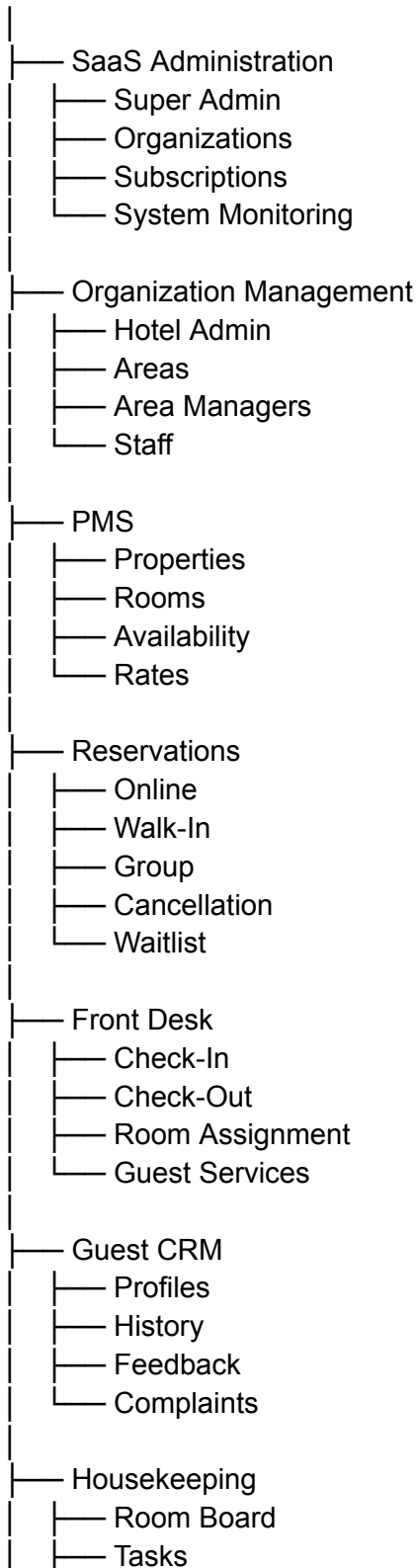
The exact development backlog should be generated from this document into:

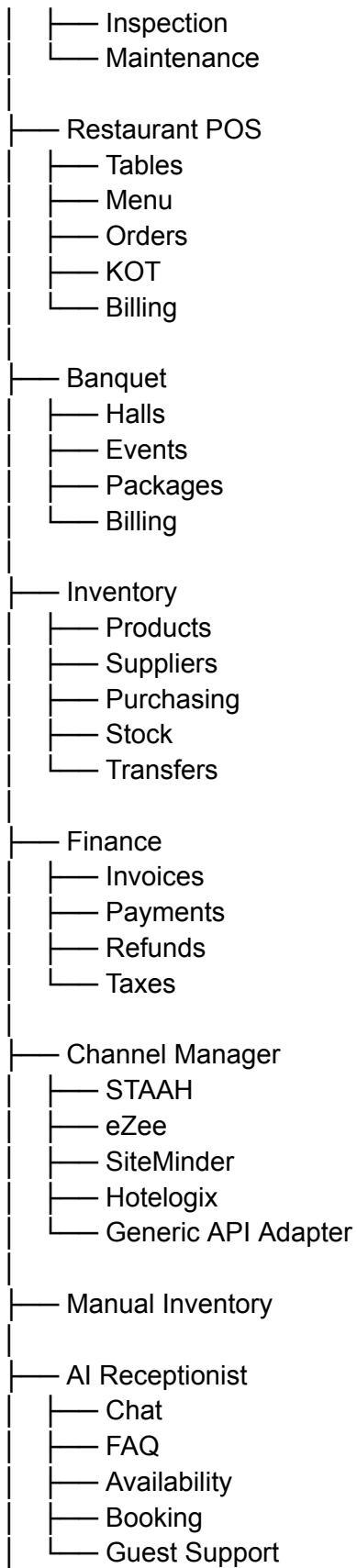
1. Epic
2. Module
3. Feature
4. Sub-feature
5. User Story
6. API
7. Database Entity
8. UI Screen
9. Acceptance Criteria
10. Test Case

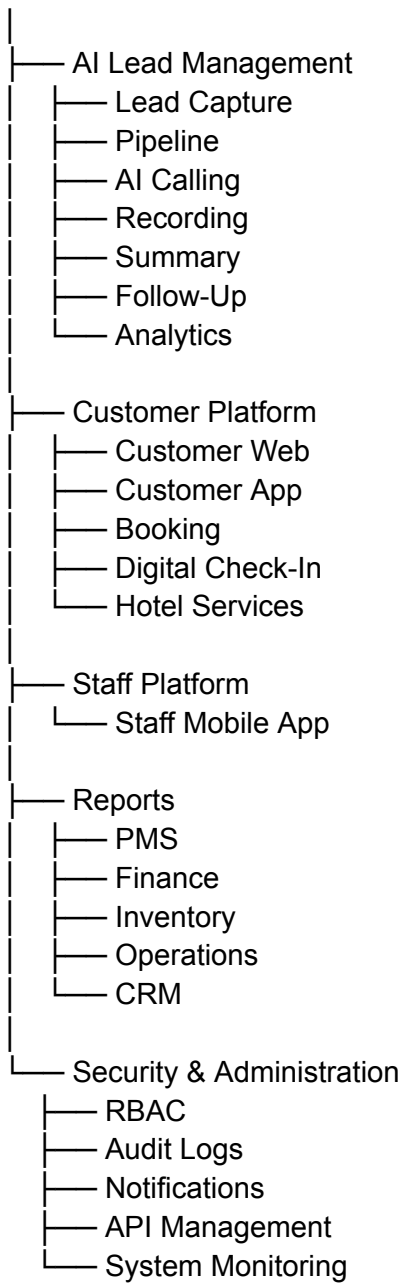
This is the correct structure for moving from an SRS into actual software development.

45. FINAL PRODUCT MODULE MAP

HOTEL OPERATING SYSTEM







46. DEVELOPMENT STANDARD

This feature document should be treated as the **Functional Master Specification**.

The next development documents should be derived from it in this order:

Master Feature List → Screen Specification → UI/UX Design → Database ERD → API Specification → Backend Architecture → Frontend Architecture → Mobile Architecture → Integration Specification → QA Test Cases → Deployment Documentation.

This prevents the development team from designing screens first and discovering later that the required permissions, database relationships, API workflows, or business rules were never defined.